WORKED SOLUTION

STEP II 1990, Question 3

Pure • Curve sketching

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q3

The key idea

Write the cubic as $f(x) = x(x^2 - 2bx + c^2)$. Two discriminants control everything: the one for $f'(x)$ tells us whether there are turning points, and the one for the quadratic factor tells us whether the curve meets the x-axis anywhere other than the origin. The seven cases are just the ways of placing b relative to the two thresholds $\frac{\sqrt{3}}{2}c$ and c. Once the cubics are drawn, $y^2 = f(x)$ comes almost for free: keep only the part where $f(x) \geq 0$, take square roots and reflect in the x-axis.

Solution

Facts that hold in every case

Let $f(x) = x^3 - 2bx^2 + c^2x = x(x^2 - 2bx + c^2)$. Then

$$f'(x) = 3x^2 - 4bx + c^2, \qquad f''(x) = 6x - 4b.$$

- *Turning points.* The discriminant of f' is $16b^2 - 12c^2 = 4(4b^2 - 3c^2)$. So f' has two distinct real zeros if $2b > c\sqrt{3}$, one repeated zero if $2b = c\sqrt{3}$, and none if $2b < c\sqrt{3}$. When there are zeros, their sum is $\frac{4}{3}b \geq 0$ and their product is $\frac{1}{3}c^2 \geq 0$, so neither is negative: all turning points lie in $x \geq 0$.
- *Roots.* $f(0) = 0$ always. The quadratic factor is $(x - b)^2 + c^2 - b^2$, so it has no real zeros if $b < c$, the double zero $x = b$ if $b = c$, and the two zeros $b \pm \sqrt{b^2 - c^2}$ if $b > c$.
- *Negative x.* For $x < 0$ each of x^3, $-2bx^2$ and c^2x is ≤ 0, and $x^3 < 0$, so $f(x) < 0$. The whole curve to the left of the y-axis lies below the x-axis.
- *Shape.* $f(x) \to \pm\infty$ as $x \to \pm\infty$; the gradient at the origin is $f'(0) = c^2$; and the only point of inflection is at $x = \frac{2}{3}b$, where f'' changes sign.

Since $c\sqrt{3} < 2c$ whenever $c > 0$, the cases run in order of increasing b for a fixed c: no turning points, then a stationary inflection, then turning points above the axis, then a touching root, then three roots.

The cubics

(i) $2b < c\sqrt{3}$. Here $f'(x) > 0$ for all x (its discriminant is negative and its leading coefficient is positive), so f is strictly increasing. Also $b < \frac{\sqrt{3}}{2}c < c$ (note $c > 0$ here), so the origin is the only root. The curve rises through the origin with gradient c^2 and has an inflection at $x = \frac{2}{3}b$.

(ii) $2b = c\sqrt{3} \neq 0$. Now $f'(x) = 3(x - \frac{2}{3}b)^2 \geq 0$, so f is still increasing, but it is momentarily level at $x = \frac{2}{3}b = c/\sqrt{3}$. This is also where $f'' = 0$, so it is a stationary point of inflection. Using $c^2 = \frac{4}{3}b^2$,

$$f\left(\tfrac{2}{3}b\right) = \tfrac{2}{3}b\left(\tfrac{4}{9}b^2 - \tfrac{4}{3}b^2 + \tfrac{4}{3}b^2\right) = \tfrac{8}{27}b^3 = \tfrac{\sqrt{3}}{9}c^3 > 0.$$

Again $b < c$, so the origin is the only root.

(iii) $c\sqrt{3} < 2b < 2c$. Now f' has two zeros $x_1 < x_2$, both positive, giving a maximum at $x_1 = \frac{1}{3}\left(2b - \sqrt{4b^2 - 3c^2}\right)$ and a minimum at $x_2 = \frac{1}{3}\left(2b + \sqrt{4b^2 - 3c^2}\right)$.

? **Ask yourself** Does the minimum dip below the x-axis?

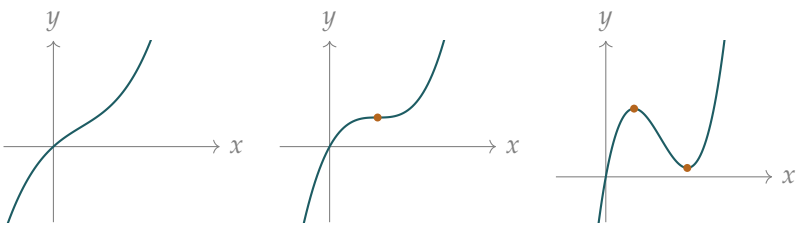
No. Since $b < c$, the factor $(x - b)^2 + c^2 - b^2$ is positive for every x, so $f(x) > 0$ for all $x > 0$. Both turning points therefore lie above the axis, and the curve still crosses the axis only at the origin. It climbs, falls back a little, then climbs for good.

(iv) $b = c \neq 0$. Now $f(x) = x(x - b)^2$. The double root at $x = b$ means the curve touches the axis there; this is the minimum, $(b, 0)$. The other zero of $f'(x) = (x - b)(3x - b)$ gives the maximum $\left(\frac{1}{3}b, \frac{4}{27}b^3\right)$.

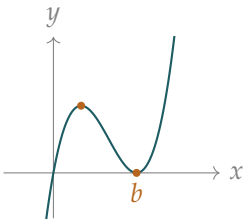
(v) $b > c > 0$. There are three roots, $0 < \alpha < \beta$, where $\alpha = b - \sqrt{b^2 - c^2}$ and $\beta = b + \sqrt{b^2 - c^2}$ (both positive because $\sqrt{b^2 - c^2} < b$). Between consecutive roots the curve must turn, so the maximum lies in $(0, \alpha)$ with $f > 0$ there and the minimum lies in (α, β) with $f < 0$ there. So $f \geq 0$ exactly on $[0, \alpha]$ and on $[\beta, \infty)$.

(vi) $c = 0$, $b \neq 0$. Now $f(x) = x^2(x - 2b)$ and $f'(x) = x(3x - 4b)$. The double root at the origin makes it a maximum where the curve touches the axis, and there is a minimum at $\left(\frac{4}{3}b, -\frac{32}{27}b^3\right)$. The curve crosses the axis at $x = 2b$.

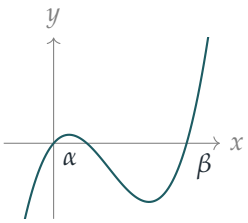
(vii) $b = c = 0$. This is $y = x^3$, with a stationary point of inflection at the origin.



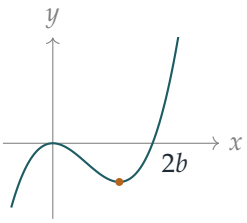
(i) (ii) (iii)



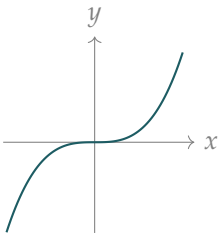
(iv)



(v)



(vi)



(vii)

The sketches use $c = 1$ with $b = \frac{1}{2}$, $\frac{\sqrt{3}}{2}$, 0.99, 1 in (i)–(iv), $(b, c) = (1, 0.8)$ in (v) and $b = 0.6$ in (vi). The vertical scales differ from panel to panel so that the small bumps in (iii) and (iv) can be seen.

The curves $y^2 = f(x)$

Three rules turn each cubic into the curve $y^2 = f(x)$:

- there are no points where $f(x) < 0$, and in particular none with $x < 0$;
- where $f(x) \geq 0$, $y = \pm\sqrt{f(x)}$, so the curve is symmetric in the x-axis and is tallest where f is largest;
- differentiating, $2y \dfrac{\mathrm{d}y}{\mathrm{d}x} = f'(x)$.

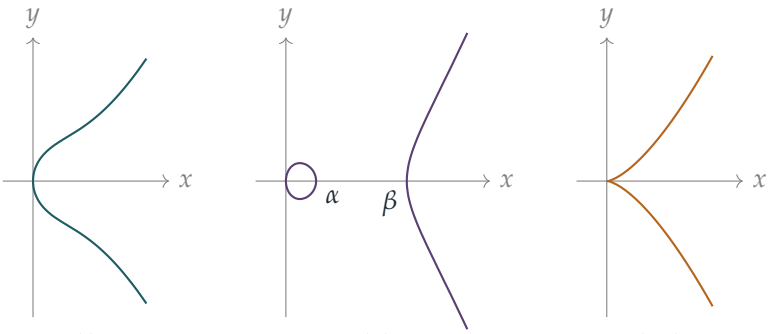
Ask yourself What does the curve look like where it meets the x-axis?

At a simple root x_0 of f we have $f'(x_0) \neq 0$, and $\dfrac{\mathrm{d}x}{\mathrm{d}y} = \dfrac{2y}{f'(x)} \to 0$ as $y \to 0$. So the tangent there is *vertical*, and the upper and lower halves join smoothly. Near such a root $y^2 \approx f'(x_0)(x - x_0)$, which is a parabola on its side. A repeated root behaves differently, as case (vii) shows.

Case (i). $f(x) \geq 0$ exactly when $x \geq 0$, and the root at 0 is simple ($f'(0) = c^2 > 0$). So the curve is a single unbounded branch. It leaves the origin with a vertical tangent, looking like $y^2 \approx c^2 x$ there, and $|y| = \sqrt{f(x)}$ increases steadily, since f is increasing, eventually growing like $x^{3/2}$. The upper half starts out concave, like $\sqrt{x}$, and ends up convex, like $x^{3/2}$, so it has a point of inflection (and, by symmetry, so does the lower half).

Case (v). $f(x) \geq 0$ on $[0, \alpha]$ and on $[\beta, \infty)$, and all three roots are simple. So the curve has two separate pieces: a closed oval through $(0,0)$ and $(\alpha,0)$, with vertical tangents at both, which is widest (in y) at the x-coordinate of the cubic's maximum; and an unbounded branch starting from $(\beta,0)$ with a vertical tangent and opening to the right like the curve in case (i). Nothing lies in the gap $\alpha < x < \beta$, where the cubic is negative.

Case (vii). $y^2 = x^3$, so $x \geq 0$ and $y = \pm x^{3/2}$. Here $\mathrm{d}y/\mathrm{d}x = \pm\frac{3}{2}x^{1/2} \to 0$ as $x \to 0$: both halves arrive at the origin *tangent to the x-axis*, from above and below, forming a sharp point called a cusp.



These use the same values of b and c as the corresponding cubics, with equal scales on the two axes.

Checks

- Put $b = c = 0$ into any general statement and it should match (vii); for example the stationary value $\frac{8}{27}b^3$ in (ii) and the minimum $-\frac{32}{27}b^3$ in (vi) both shrink to the origin.
- With $c = 1$ and $b = 0.99$ (case (iii)) the turning points are at about $(0.340, 0.150)$ and $(0.980, 0.020)$, both above the axis as claimed. With $(b, c) = (1, 0.8)$ the roots are 0, 0.4, 1.6 and the minimum is about -0.388 at $x \approx 1.147$, inside $(0.4, 1.6)$.
- A numerical test over thousands of random (b, c) in cases (i), (iii) and (v) confirmed the number and position of the turning points and roots, and the set where $f \geq 0$.

Common slips

- Drawing the minimum in case (iii) below the axis: with $b < c$ the cubic is positive for all $x > 0$.
- Drawing $y^2 = f(x)$ with a corner, or a horizontal tangent, where it crosses the x-axis at a simple root. The tangent there is vertical.
- Forgetting that $y^2 = f(x)$ has no points where the cubic is negative, so in case (v) the oval and the branch are disconnected.

Beyond STEP

A curve $y^2 = x^3 + Ax^2 + Bx + C$ whose cubic has distinct roots is an *elliptic curve*, one of the central objects of modern number theory. Case (v), with three real roots, gives the two-piece picture of an oval plus a branch; case (i), with one real root, gives one piece. When roots coincide the curve becomes singular: case (vii) has a cusp, and case (iv) would give a curve that crosses itself at $(b, 0)$. For an introduction to the remarkable way points on such curves can be "added", see Silverman and Tate, *Rational Points on Elliptic Curves*.

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